



The Earth — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Chapter 01_vi_The_Earth

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Earth ki Shape aur Discovery

- **Aristotle** sabse pehle bataya ki Earth spherical hai ('On the Heavens' mein). [Q1]
- Life ke context mein Earth ko '**Green Planet**' kehte hain (vegetation/life ki widespread presence ki wajah se). [Q2]
- Earth ka diameter ~**12,800 km** hai (equatorial ~12,756 km, polar ~12,714 km). [Q3]
- Earth ka shape **oblate spheroid** / **geoid** hai — perfect sphere nahi (Equator pe bulge, poles pe flat). [Q4]
- Weight Equator se poles ki taraf badhta hai (centrifugal effect Equator pe max) — high latitude jaane pe weight **badhta** hai, ghatta nahi. [Q4]



EXAM TRAP

- Assertion 'weight decreases with higher latitude' **FALSE** hai — actually weight **INCREASES**; sirf Reason (Earth not perfect sphere) **TRUE** hai — 'A false, R true' pattern. [Q4]

Earth ka Structure — Core, Crust

- Earth's crust mein sabse abundant element **Oxygen** hai (46.60%), phir Silicon (27.72%), Aluminium (8.13%). [Q6]
- Earth ka core mainly **Nickel aur Iron (NiFe)** se bana hai — outer core liquid, inner core solid. [Q27, Q28]
- Outer core ~2,260 km thick (liquid Fe+Ni); inner core radius ~1,220 km, extremely hot/dense, mainly Iron. [Q28]
- Moon ka core bhi solid inner + liquid outer hota hai (Earth jaisa hi structure, lekin proportion mein chhota). [Q5]

Earth ki Motion — Rotation, Revolution, Speed

- Sunlight Sun se Earth tak pahunchne mein ~**8 minutes 19 seconds** leta hai. [Q7, Q8]
- Sun Earth ka nearest star hai (~149.6 million km). [Q9]

- Mars given options mein Earth ke sabse close hai (long-term); Venus closest-approach mein sabse close aa sakta hai (favourable alignment mein). [Q10,Q11]
- Earth 1 revolution ~**365.25 days** (365 din 6 ghante 9 min) mein complete karti hai — isko Solar Year kehte hain. [Q12]
- Earth ki average orbital speed ~**30 km/sec** (29.8 km/s) hai. [Q13]
- Earth apni axis pe rotate karte hue ~**28 km/minute** distance cover karti hai (equatorial rotation speed ~1,670 km/hr). [Q14]
- Earth ki equatorial circumference ~**40,000 km** hai. [Q15]

Earth ki Axis Tilt aur Seasons

- Earth ka rotational axis ecliptic plane ke normal se ~**23.5°** tilted hai. [Q16,Q31]
- Seasonal changes 2 factors se hote hain: **Axial tilt (23½°)** + **Revolution around Sun** — dono zaroori hain. [Q21,Q22]
- Day-Night Earth ke **rotation** se hote hain (revolution se nahi). [Q19]
- 21 March/23 Sept ko Sun ki rays **Equator** pe vertically padti hain — day-night equal hote hain. [Q20]
- Delhi mein summer noon temperature winter se zyada hota hai kyunki winter mein sunlight zyada **obliquely** (tirchhi) padti hai. [Q23]

Magnetic Field aur Poles

- Statement: Magnetic field axis 23½° tilted (Geographic axis se) — **INCORRECT** hai; Magnetic North Pole northern Canada peninsula pe hai aur Magnetic Equator Thumba (South India) se guzarti hai — dono **CORRECT**. [Q17]
- Earth ka magnetic field har few hundred thousand years mein reverse hota raha hai (geological history mein). [Q32]
- North Pole = Earth ke axis ka northern end; 45° latitude ki length Equator ki half NAHI hoti (misconception); Earth's axis **parallelism** maintain karta hai revolution ke dauraan. [Q18]
- Aphelion (~4 July) pe Earth ki orbital speed **kam** hoti hai, zyada nahi. [Q18]

Satellites aur Orbital Mechanics

- Agar Sun-Earth distance ¼ kam ho jaaye, toh year ki duration **kam** ho jaayegi (smaller orbit = faster revolution). [Q24]
- Assertion: satellites hamesha eastward launch hote hain — **FALSE** (polar-orbit satellites north/south bhi jaate hain); Reason: west-to-east rotation launch-velocity mein help karta hai — **TRUE**. [Q25]
- Agar ball ko satellite se release kiya jaaye bina extra velocity, wo **same orbit mein Earth ki parikrama continue** karegi (satellite jaisi hi speed se). [Q26]

Earth ka History aur Atmosphere

- Statement check: Magnetic field reversal **CORRECT**; early atmosphere mein 54% oxygen tha — **INCORRECT** (early atmosphere mein bahut kam/negligible free oxygen tha); living organisms ne early atmosphere modify kiya — **CORRECT** (photosynthetic algae/cyanobacteria). [Q32]

- Aryabhata sabse pehle explain kiya ki Earth ka apne axis pe rotation hi Sun ke daily rise/set ka apparent karan hai. [Q33]

Deserts aur Earth Day

- Desert formation ki probability ~**23° latitude** (subtropical belt, 15°-30°) ke around zyada hoti hai — subtropical high-pressure descending dry air ki wajah se. [Q34]
- **World/International Earth Day** har saal **22 April** ko manaya jaata hai (first Earth Day 1970 mein). [Q29, Q30]